WORKED SOLUTION

STEP I 1995, Question 2

Pure • Integration by Substitution

> **STEP I 1995, Question 2**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/95-s1-q2

The key idea

In part (i) neither integral is easy on its own, but they pair up beautifully: adding them gives a fraction that cancels to 1, and subtracting them gives a numerator that is exactly the derivative of the denominator. Two easy integrals give two simultaneous equations for S and T. In part (ii) the substitution $x = \sin^2 t$ is chosen because it turns $\frac{1}{x} - 1$ into a perfect square, $\cot^2 t$, so the square root disappears; after that it is double-angle formulae all the way.

Solution

Part (i)

Adding the integrands,

$$S + T = \int \frac{\cos x + \sin x}{\cos x + \sin x}\,\mathrm{d}x = \int 1\,\mathrm{d}x = x + c_1.$$

Subtracting them,

$$S - T = \int \frac{\cos x - \sin x}{\cos x + \sin x}\,\mathrm{d}x.$$

The numerator is the derivative of the denominator, since $\frac{\mathrm{d}}{\mathrm{d}x}(\cos x + \sin x) = -\sin x + \cos x$. So this is of the form $\int \frac{f'(x)}{f(x)}\,\mathrm{d}x$, and

$$S - T = \ln|\cos x + \sin x| + c_2.$$

Adding and subtracting these two equations and halving,

$$S = \tfrac{1}{2}(x + \ln|\cos x + \sin x|) + C,$$
$$T = \tfrac{1}{2}(x - \ln|\cos x + \sin x|) + C',$$

where C and C' are arbitrary constants. These hold on any interval on which $\cos x + \sin x \neq 0$, where the integrands are defined.

Part (ii)

Let I be the required integral. With $x = \sin^2 t$ we have $\mathrm{d}x = 2\sin t\cos t\,\mathrm{d}t$. For the limits, $x = \frac{1}{4}$ gives $\sin t = \frac{1}{2}$, so $t = \frac{\pi}{6}$, and $x = \frac{1}{2}$ gives $\sin t = \frac{1}{\sqrt{2}}$, so $t = \frac{\pi}{4}$. (We take t in $[\frac{\pi}{6}, \frac{\pi}{4}]$; as t runs over this interval, x increases from $\frac{1}{4}$ to $\frac{1}{2}$.)

Next, the square root:

$$\frac{1}{x} - 1 = \frac{1-\sin^2 t}{\sin^2 t} = \frac{\cos^2 t}{\sin^2 t}, \qquad \text{so} \qquad \sqrt{\frac{1}{x}-1} = \frac{\cos t}{\sin t}.$$

? **Ask yourself** Why is the square root $\cos t/\sin t$ and not $|\cos t/\sin t|$?

In general $\sqrt{\cot^2 t} = |\cot t|$. But for $\frac{\pi}{6} \le t \le \frac{\pi}{4}$ both $\sin t$ and $\cos t$ are positive, so $\cot t > 0$ and the modulus can be dropped. This is why we chose t in the first quadrant.

Substituting, the $\sin t$ cancels:

$$I = \int_{\pi/6}^{\pi/4} (1 - 4\sin^2 t)\,\frac{\cos t}{\sin t}\,2\sin t\cos t\,\mathrm{d}t = \int_{\pi/6}^{\pi/4} 2\cos^2 t\,(1-4\sin^2 t)\,\mathrm{d}t.$$

Now we use double angles to make this easy to integrate. Since $2\cos^2 t = 1 + \cos 2t$ and $8\sin^2 t\cos^2 t = 2\sin^2 2t = 1 - \cos 4t$,

$$2\cos^2 t - 8\sin^2 t\cos^2 t = (1+\cos 2t) - (1-\cos 4t) = \cos 2t + \cos 4t.$$

Therefore

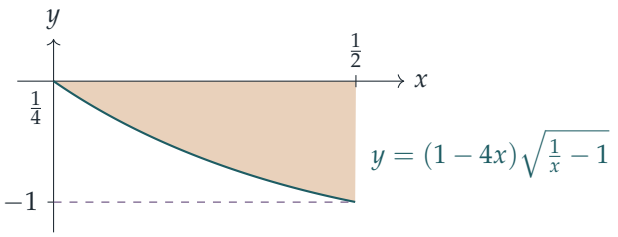
$$\begin{aligned} I &= \int_{\pi/6}^{\pi/4} (\cos 2t + \cos 4t)\,\mathrm{d}t = \left[\tfrac{1}{2}\sin 2t + \tfrac{1}{4}\sin 4t\right]_{\pi/6}^{\pi/4} \\ &= \left(\tfrac{1}{2}\sin\tfrac{\pi}{2} + \tfrac{1}{4}\sin\pi\right) - \left(\tfrac{1}{2}\sin\tfrac{\pi}{3} + \tfrac{1}{4}\sin\tfrac{2\pi}{3}\right) \\ &= \tfrac{1}{2} - \left(\tfrac{\sqrt{3}}{4} + \tfrac{\sqrt{3}}{8}\right), \end{aligned}$$

so

$$\int_{1/4}^{1/2} (1-4x)\sqrt{\frac{1}{x}-1}\,\mathrm{d}x = \frac{1}{2} - \frac{3\sqrt{3}}{8} = \frac{4-3\sqrt{3}}{8}.$$

Checks

- Sign: on $(\frac{1}{4}, \frac{1}{2}]$ we have $1 - 4x < 0$ and the square root is positive, so the integrand is negative and I must be negative. Indeed $4 - 3\sqrt{3} \approx -1.196$, so $I \approx -0.1495$. The graph below shows the integrand.
- Simpson's rule applied directly to the original integral gives $-0.149519\ldots$, matching $\frac{1}{2} - \frac{3\sqrt{3}}{8}$ to ten decimal places.
- For part (i), differentiate: $\frac{\mathrm{d}}{\mathrm{d}x}\,\frac{1}{2}\big(x + \ln(\cos x + \sin x)\big) = \frac{1}{2}\left(1 + \frac{\cos x - \sin x}{\cos x + \sin x}\right) = \frac{\cos x}{\cos x + \sin x}$. Over $[0, 1]$, for example, both S and T agree numerically with quadrature (0.66168 and 0.33832).



Common slips

- Forgetting to change the limits to $t = \frac{\pi}{6}$ and $t = \frac{\pi}{4}$, or substituting back to x incorrectly.
- Dropping the factor $2\sin t\cos t$ from $\mathrm{d}x$, or writing $\mathrm{d}x = 2\sin t\,\mathrm{d}t$.
- Giving a positive answer: the integrand is negative throughout, so a positive value signals a sign error.

Beyond STEP

The trick in part (i), pairing an awkward integral with a "partner" so that the sum and difference are both easy, appears again and again, for instance with $\int \mathrm{e}^x \cos x\,\mathrm{d}x$ and $\int \mathrm{e}^x \sin x\,\mathrm{d}x$, and in its complex form by integrating $\mathrm{e}^{(1+\mathrm{i})x}$. Part (ii) is a small example of a general principle: a well-chosen substitution can turn an algebraic integrand into a trigonometric polynomial, which is the starting point of Fourier series. For many more integrals of this flavour, with careful discussion, see Siklos, *Advanced Problems in Mathematics*.

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